Vitto AI-First Credit Infrastructure

The objective of this project was to present Vitto as a serious infrastructure layer for Banks and NBFCs rather than a typical SaaS product. Instead of designing a visually heavy or marketing-driven website, I focused on making the system feel credible to someone evaluating it from a technical or operational perspective. This led to a restrained design approach with a dark theme and minimal use of color, aiming to reflect stability and clarity rather than visual noise.

The entire website was structured around one central idea: Vitto is a decisioning system, not just a set of tools. This influenced how each page was designed. The homepage highlights the limitations of existing lending systems, such as fragmentation and static rule engines. The platform page explains how decisions are actually made using data, models, and policy. The lifecycle page organizes the lending process into layers, showing how different components connect instead of presenting them as isolated features. The focus throughout was on system behavior rather than feature listing.

On the technical side, I followed the required stack using React for the frontend and Node.js with Express for the backend. PostgreSQL was used for structured lead data, while MongoDB was used for OTP sessions with a time-to-live configuration. A key design decision was separating temporary and persistent data. OTPs are short-lived and suited for automatic expiry, whereas leads are long-term structured records and require relational consistency. This separation aligns with how production systems typically handle data.

The most challenging part of the assignment was structuring the lending lifecycle in a way that reflects real system behavior. Simply listing modules like KYC or collections does not explain how they interact. To address this, I grouped the system into five layers: acquisition, underwriting and loan origination, loan management, collections, and CRM. For each layer, I defined inputs, outputs, and its role in the overall flow. This helped convert a feature list into a connected system with clear data movement.

If I had more time, I would focus on making the system more interactive. I would add a visual decision flow to show how data moves through models and rule engines and build small interactive demos for underwriting or collections logic. I would also expand the API layer with live request-response examples. These additions would make the platform easier to understand in practice, not just in theory.

Thought Leadership Article: Retrofit AI vs AI-Native Infrastructure in BFSI (Deliverable 5)

AI adoption in financial services is accelerating, but most implementations are being driven by urgency rather than system design. Instead of rebuilding for AI, institutions are layering AI onto existing systems such as LOS, LMS, and CRM. While this works initially, it creates limitations that become visible in decision quality, speed, and maintainability. Two approaches are emerging. Retrofit AI adds intelligence on top of legacy systems. AI-native infrastructure builds systems where data, models, and policy operate together from the start. The difference lies in how decisions are produced.

In retrofit setups, data is fragmented across systems with different structures. AI models must combine this data through intermediate layers, which introduces delays and inconsistencies. Over time, integration becomes complex, and maintenance becomes harder as changes in one system affect others. More importantly, decision quality suffers. Models depend on complete and timely data, but fragmented systems provide only partial context. As a result, decisions are slower and less accurate.

AI-native systems address this by integrating data ingestion, model inference, and rule execution into a single pipeline. This allows decisions to be made in real time with full context. Explainability is another key difference. In retrofit systems, decision logic is spread across multiple layers, making it difficult to trace outcomes. In AI-native systems, every decision is directly linked to input data, model output, and policy rules, making the system easier to audit and more reliable for compliance.

Ultimately, traditional lending platforms are designed to process transactions, moving applications from one step to another. AI-native systems are designed to determine what should happen at each step based on data and policy. Institutions can continue extending legacy systems or move toward architectures where decision-making is central. Long-term efficiency and accuracy depend on that choice.

By – Divye Bisaria